Task 2 Numerical Methods

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PROBLEM 1

Traits	Bisection	Newton	Secant
Rate of Convergence	Linear	Quadratic	Superlinear
Per Iteration Cost	Low (Simple operations)	Moderate-High (Requires derivative)	Moderate (Requires two previous points)
Global Convergence	Yes, if initial brackets are chosen properly	No, only local	No, only local
Extends to Higher Dimensions	No (But there are multidimensional equivalents)	Yes (Newton's method for systems of equations)	No (But there are multidimensional equivalents)
Conditions for Convergence	Function must be continuous and different signs at endpoints	Function must be smooth and close to the root initially	Function must be smooth and two initial approximations are required

PROBLEM 2

3.1.2

a) According to the content on page 78 , the width of the interval at the n'th step is given by:

$$b_n - a_n = 2^{-n}\,(b_0 - a_0)$$

For the initial interval $[1.5, 3.5]$, $b_0 = 3.5$ and $a_0 = 1.5$. So, $b_0 - a_0 = 3.5 - 1.5 = 2$. Therefore, the width of the interval at the n'th step would be:

$$b_n - a_n = 2^{-n} \times 2 = 2^{-n+1}$$

b) The maximum distance between the root r and the midpoint c_n of the interval $[a_n, b_n]$ is given by:

$$|r - c_n| \leq \frac{1}{2}\,(b_n - a_n)$$

Using the expression for $b_n - a_n$ I derived before, this inequality becomes:

$$|r - c_n| \leq \frac{1}{2} \times 2^{-n+1} = 2^{-n}$$

Thus, the maximum distance possible between the root r and the midpoint of the interval at the n'th step is 2^{-n}.

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3.1.9

For the case $[a_0, b_0] = [0, 1]$, the root r in binary representation can be expressed as $r = 0.d_1d_2d_3\ldots$ In decimal form, $r = \sum_{n=1}^{\infty} \frac{d_n}{2^n}$. For a general interval $[a_0, b_0]$, the root r is:

$$r = a_0 + (b_0 - a_0) \sum_{i=1}^{\infty} \frac{d_i}{2^i}$$

Here $d_n = 0$ if the interval at the n-th step is the left half of the previous interval, and $d_n = 1$ otherwise.

3.2.9

To solve for x_2 using Newton's method, I first need to find the derivative $f'(x)$ of $f(x) = x^3 - 2$. The derivative is $f'(x) = 3x^2$. Newton's method uses the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Starting with $x_0 = 1$, let's find x_1 and then x_2. To find x_1 :

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$
$$x_1 = 1 - \frac{(1)^3 - 2}{3(1)^2}$$
$$x_1 = 1 - \frac{-1}{3}$$
$$x_1 = 1 + \frac{1}{3}$$
$$x_1 = \frac{4}{3}$$

To find x_2 :

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$
$$x_2 = \frac{4}{3} - \frac{\frac{64}{27} - \frac{54}{27}}{3 \cdot \frac{16}{9}}$$
$$x_2 = \frac{4}{3} - \frac{10}{48}$$
$$x_2 = \frac{96}{72} - \frac{15}{72}$$
$$x_2 = \frac{9}{8}$$

$$x_2 = \frac{4}{3} - \frac{\left(\frac{4}{3}\right)^3 - 2}{3\left(\frac{4}{3}\right)^2}$$
$$x_2 = \frac{4}{3} - \frac{\frac{10}{27}}{\frac{48}{27}}$$
$$x_2 = \frac{4}{3} - \frac{5}{24}$$
$$x_2 = \frac{81}{72}$$

Thus, $x_2 = \frac{9}{8}$.